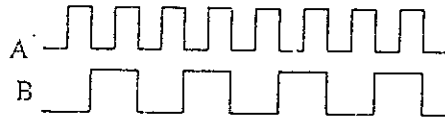
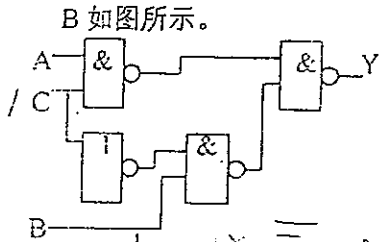


一、图中，当控制端 C=1 和 C=0 两种情况时，分别求输出 Y 的表达式和波形。输入波形 A、B 如图所示。

$A=1$
 $A=0$



C=1 $\rightarrow$ A-1

$Y = \overline{AC} \cdot \overline{CB}$
C=1 $\rightarrow$ $Y = \overline{A} = A$
C=0 $\rightarrow$ $Y = \overline{B} = B$

$Y_1 = AB + \overline{A}\overline{B} + \overline{A}\overline{B}$

二、用逻辑代数运算法则化简下列各式： $Y_2 = (AB + \overline{A}\overline{B} + \overline{A}B)(A + B + D + \overline{A}\overline{B}\overline{D})$

$Y_1 = AB + \overline{A}\overline{B} + \overline{A}B$
 $= A(B + \overline{B}) + \overline{A}B$
 $= A + \overline{A}B$
 $= A + B$

$Y_3 = ABC + \overline{A} + \overline{B} + \overline{C} + D$

$Y_3 = ABC + \overline{A} + \overline{B} + \overline{C} + D$
 $= ABC + \overline{A} + \overline{B} + \overline{C} + D$
 $= ABC + \overline{A} + \overline{B} + \overline{C} + D$
 $= 0 + D$
 $= D$

$Y = (AB + \overline{A}\overline{B} + \overline{A}B)(A + B + D + \overline{A}\overline{B}\overline{D})$
 $= (A + \overline{A}B)(A + B + D + \overline{A}\overline{B}\overline{D})$
 $= (A + \overline{A}B)(A + B + D + \overline{A}\overline{B}\overline{D})$
 $= A + \overline{A}B + AD + \overline{A}B\overline{D} + AB + B^2 + BD + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D}$
 $= A + \overline{A}B + AD + \overline{A}B\overline{D} + AB + B + BD + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D}$
 $= A + \overline{A}B + AD + \overline{A}B\overline{D} + AB + B + BD + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D}$
 $= A + \overline{A}B + AD + \overline{A}B\overline{D} + AB + B + BD + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D} + \overline{A}B\overline{D}$

$Y_1 = AB + \overline{A}\overline{B} + \overline{A}B$

三、用卡诺图化简下列逻辑函数： $Y_2 = \overline{A}B\overline{C}D + \overline{A}BCD + \overline{A}BC\overline{D} + \overline{A}B\overline{C}\overline{D}$

$Y_3 = A + \overline{A}\overline{B} + \overline{A}\overline{B}C + \overline{A}\overline{B}\overline{C}D$

AB \ CD	00	01	11	10
00				
01				
11				
10				

$Y_1 = B$

相同的留下
不同的去掉

AB \ CD	00	01	11	10
00				
01				
11				
10				

$Y_2 = \overline{A}B\overline{C}D + \overline{A}BCD$

AB \ CD	00	01	11	10
00				
01				
11				
10				

$Y_3 = \overline{A}\overline{B}\overline{C}\overline{D} = A + B + C + D$

四、化简 $Y = AD + \overline{C}D + \overline{A}C + \overline{B}C + DC$ 并用 74LS20 双 4 输入与非门组成电路。

$Y = AD + \overline{C}D + \overline{A}C + \overline{B}C + DC$
 $= AD + \overline{C}(D + \overline{D}) + \overline{A}C + \overline{B}C$
 $= AD + \overline{C} + \overline{A}C + \overline{B}C$
 $= AD + (1 + \overline{A} + \overline{B})\overline{C}$
 $= AD + \overline{C}$

14	13	12	11	10	9	8
74LS20						
1	2	3	4	5	6	7

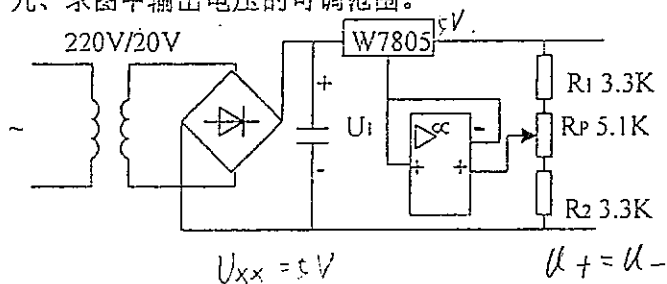
AB \ CD	00	01	11	10
00				
01				
11				
10				

$Y = AD + \overline{C}$

八、要求负载电压 $U_o=30V$ ；电流 $I_o=150mA$ ；采用单相桥式整流，带电容滤波。已知交流频率为 $50Hz$ ，选择管子的型号和滤波电容，并与单相半波整流电路比较，带电容滤波后管子承受的反向电压是否相同？

$$11) \quad I_D = \frac{1}{2} I_o = 75mA \quad U_{RM} = \sqrt{2} U = \sqrt{2} \frac{U_o}{1.2}$$

九、求图中输出电压的可调范围。



$$U_{o \min} = \frac{R_1 + R_p + R_2}{R_1 + R_p} U_{xx} = 6.96V$$

$$U_{o \max} = \frac{R_1 + R_p + R_2}{R_1} U_{xx} = 12.77V$$

$$R_F = \infty: U_o = (1 + \frac{R_F}{R_1}) U_i = U_i$$

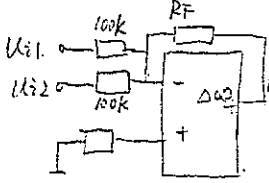
五、按下列运算关系式画出运算电路，并求电阻值。

(1) $u_{o1} = -(u_{i1} + 0.2u_{i2})$ ($R_F = 100K$)

(2) $u_{o2} = 2u_{i2} - u_{i1}$ ($R_F = 10K$)

(3) $u_{o3} = -10 \int u_{i1} dt - 5 \int u_{i2} dt$ ($C_F = 1\mu F$)

(4) $u_{o4} = 0.5u_{i1}$ ($R_F = 10K$)



$$-\frac{R_F}{R_x} u_{o1} = \frac{R}{10} (u_{i1} + 0.2u_{i2})$$

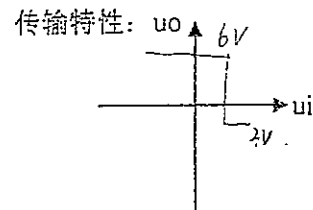
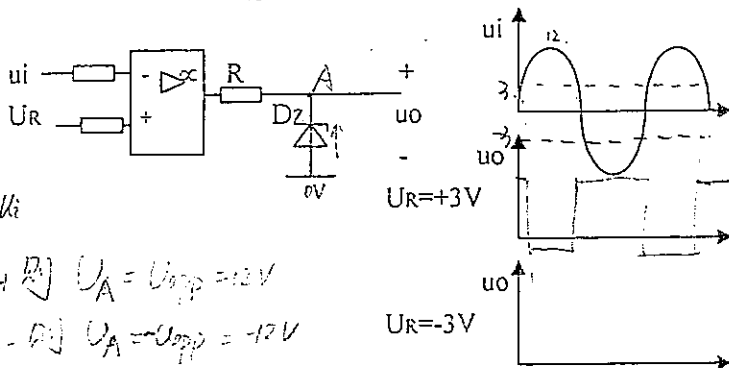
$$R_x = 1 \quad 1.8 \quad \frac{100}{2.5} = 40$$

$$(1 + \frac{R_F}{R_x}) u_{i2}$$

$$-\frac{1}{C R_F} \int u_{i2} dt = u_{i2} = -u_{i1}$$

$$u_{i1} = 2u_{i2}$$

六、图中，运放最大输出电压 $U_{OPP} = \pm 12V$ ，稳压管稳定电压 $U_Z = 6V$ ，正向压降 $U_D = 0.7V$ ， $u_i = 12\sin\omega t V$ 。当参考电压 $U_R = +3V$ 和 $-3V$ 两种情况下，画出传输特性和输出电压波形。



$$u_o = -u_i$$

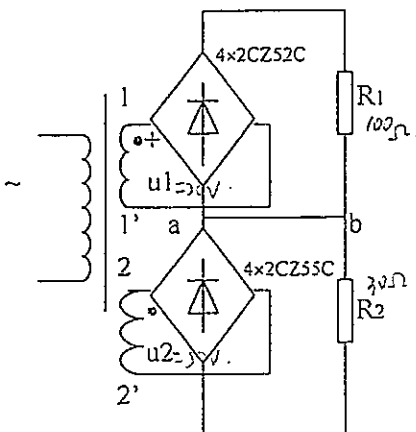
$$u_i > u_R \text{ 时 } u_A = U_{OPP} = 12V$$

$$u_i < u_R \text{ 时 } u_A = -U_{OPP} = -12V$$

$$\text{当 } U_R = 3V \text{ 时 } u_i > 3V \text{ 时 } u_A = -12V, u_o = -0.7V$$

$$u_i \leq 3V \text{ 时 } u_A = +12V, u_o = 6V$$

七、图示电路，变压器二次侧电压有效值 $U_1 = 20V$ ， $U_2 = 50V$ ； $R_1 = 100\Omega$ ， $R_2 = 30\Omega$ ，二极管的参数如表所示。(1) 校验图中的二极管是否合适；(2) 若将 2-2 的极性接反，对电路有无影响；(3) 若将 a-b 间的连线去掉，电路是否仍能正常工作？此时输出电压 U_o 和输出电流 I_o 等于多少？所选二极管是否仍然合适？



型号	IOM/A	URWM/V
2CZ52C	0.1	100 适合
2CZ55C	1	100 不适合

$$U_{o1} = 0.9 U_1 = 18V$$

$$U_{o2} = 0.9 U_2 = 45V$$

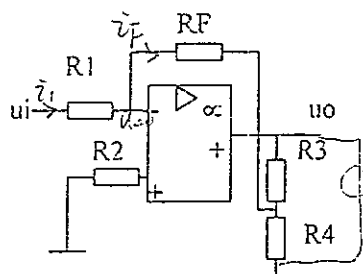
$$I_{o1} = \frac{1}{2} \frac{U_{o1}}{R_1} = \frac{1}{2} \times \frac{18}{100} = 0.09A$$

$$I_{o2} = \frac{1}{2} \frac{U_{o2}}{R_2} = \frac{1}{2} \times \frac{45}{30} = 0.75A$$

$$U_{RM1} = \sqrt{2} U_1 = 28.28V$$

$$U_{RM2} = \sqrt{2} U_2 = 70.71V$$

一、图示电路, $R_F \gg R_4$; 证明: $A_{uf} = \frac{u_o}{u_i} = -\frac{R_F}{R_1} (1 + \frac{R_3}{R_4})$



$$u_- \approx u_+ = 0V \quad i_1 = \frac{u_i}{R_1} \quad i_f \approx i_F$$

$$u_o - u_- = -i_F R_F + \frac{R_3}{R_F // R_4 + R_3} u_o$$

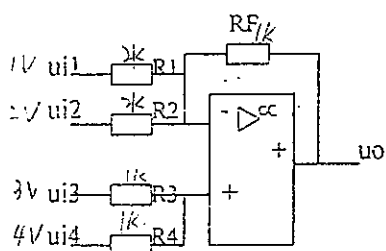
$$\because R_F \gg R_4 \therefore R_F // R_4 \approx R_4$$

$$u_o = -\frac{R_F}{R_1} u_i + \frac{R_3}{R_4 + R_3} u_o$$

$$u_o = -\frac{R_F}{R_1} u_i \frac{R_4 + R_3}{R_4}$$

$$\therefore A_{uf} = \frac{u_o}{u_i} = -\frac{R_F}{R_1} (1 + \frac{R_3}{R_4})$$

二、已知 $u_{i1}=1V$; $u_{i2}=2V$; $u_{i3}=3V$; $u_{i4}=4V$; $R_1=R_2=2K$; $R_3=R_4=R_F=1K$, 求输出电压 u_o 。



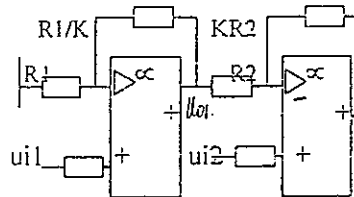
$$u_o = -(\frac{R_F}{R_1} u_{i1} + \frac{R_F}{R_2} u_{i2}) + (1 + \frac{R_F}{R_1}) (\frac{R_4}{R_3 + R_4} u_{i3} + \frac{R_3}{R_3 + R_4} u_{i4})$$

$$= -(\frac{1}{2} \times 1 + \frac{1}{2} \times 2) + (1 + 1) (\frac{1}{1+1} \times 3 + \frac{1}{1+1} \times 4)$$

$$= 5.5V$$

$$R_1' = R_1 // R_2 = 1K\Omega$$

三、求输出电压 u_o 与各输入电压的关系式。



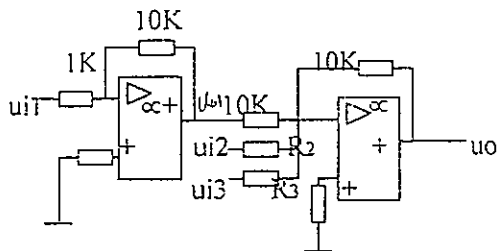
$$R_F = R_1 \quad u_{o1} = (1 + \frac{R_F}{R_1}) u_{i1} = (1 + 1) u_{i1}$$

$$u_o = -\frac{R_F}{R_2} u_{o1} + (1 + \frac{R_F}{R_2}) u_{i2}$$

$$= (1 + 1) u_{o1} + (1 + 1) u_{i2}$$

$$= (u_{i2} - u_{i1}) (1 + 1)$$

四、求输出电压 u_o 与各输入电压的关系式。 $R_2=5K$; $R_3=2K$ 。



$$u_{o1} = -\frac{R_F}{R_1} u_{i1} = -\frac{10K\Omega}{1K\Omega} u_{i1} = -10u_{i1}$$

$$u_o = -(\frac{R_F}{R_1} u_{o1} + \frac{R_F}{R_2} u_{i2} + \frac{R_F}{R_3} u_{i3})$$

$$= -(\frac{10}{10} (-10u_{i1}) + \frac{10}{5} u_{i2} + \frac{10}{2} u_{i3})$$

$$= 10u_{i1} - 2(u_{i2} - 5u_{i3})$$

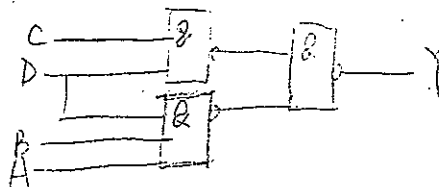
五、某同学参加四门课程考试，规定如下：

- (1) 课程 A 及格得 1 分，不及格得 0 分
- (2) 课程 B 及格得 2 分，不及格得 0 分
- (3) 课程 C 及格得 4 分，不及格得 0 分
- (4) 课程 D 及格得 5 分，不及格得 0 分

若总分大于等于 8 分可以结业，用与非门画出实现上述要求的逻辑电路。

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	1	1	0
10	0	0	1	0

$$Y = CD + ABD = \overline{CD} \overline{ABD}$$

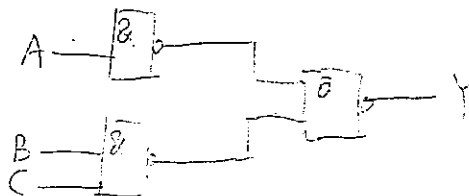


六、某汽车驾驶员培训班结业考试，有三名评判员，其中 A 是主评判员，B、C 为副评判员。

在评判时按少数服从多数的原则通过，但主评判员认为合格，亦可通过。用与非门构成逻辑电路实现此评判。合格为“1”，不合格为“0”。

A \ BC	00	01	11	10
0	0	0	1	0
1	1	1	1	1

$$Y = BC + A = \overline{A \cdot \overline{BC}}$$



七、*用 74LS153 双 4 选 1 数据选择器来实现全加器。

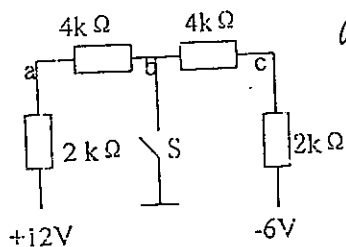
直流电路 选择填空

班级

姓名

学号

1. 求图示电路中开关S闭合和断开两种情况下a、b、c三点的电位。



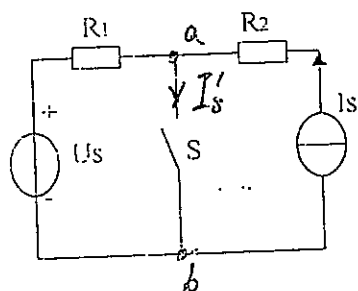
a: S 闭合 $V_b = 0$ $V_a = \frac{12 \times 4}{2+4} = 8V$ $V_c = \frac{-6}{2+4} \times 4 = -4V$

b: S 断开: $I_{abc} = \frac{12 - (-6)}{2+4+4+2} = \frac{3}{2} mA$

$V_c = 2 \times \frac{3}{2} + (-6) = -3V$ $V_b = 4 \times \frac{3}{2} + V_c = 3V$

$V_a = \frac{3}{2} \times 4 + V_b = 6 + 3 = 9V$

2. 求图示电路中, 已知 $U_s = 6V$, $I_s = 2A$, $R_1 = R_2 = 4\Omega$ 。求: 开关S闭合时通过开关的电流和开关S断开时两端的电压 (在图中注明所选择的参考方向)。

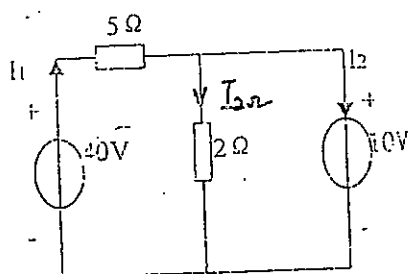


$I'_s = \frac{U_s}{R_1} + I_s = \frac{6}{4} + 2 = 3.5A$

$U_{ab} = I_s R_1 + U_s = 2 \times 4 + 6 = 14V$

$U_{ab} = U_s - (I_s R_1) = 14V$

3. 求图中通过恒压源的电流 I_1 、 I_2 及其功率, 并说明是起电源作用还是负载作用。



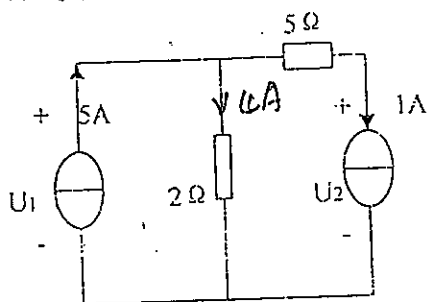
$I_1 = \frac{40-10}{5} = 6A$ $I_{2n} = \frac{10}{2} = 5A$

$I_2 = I_1 - I_{2n} = 6 - 5 = 1A$

$\therefore 40V$ 电压源 $\Rightarrow$ 电源

$10V$ 电压源 $\Rightarrow$ 负载

4. 求图中恒流源两端的电压 U_1 、 U_2 及其功率, 并说明是起电源作用还是负载作用。



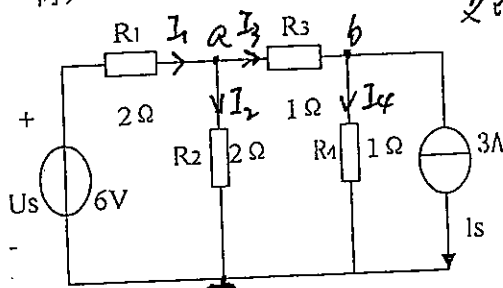
$U_1 = 2 \times 4 = 8V$

$U_2 = 8 - 5 \times 1 = 3V$

$5A$ 电流源 $\Rightarrow$ 电源

$1A$ 电流源 $\Rightarrow$ 负载

5. 用支路电流法和节点电压法分别求解下图中各支路电流 (在图中注明所选择的参考方向)



支路电流法:

$$\begin{cases} I_1 - I_2 - I_3 = 0 \\ I_3 - I_4 - I_5 = 0 \\ 2I_1 + 2I_2 = V_s \\ I_3 + I_4 - 2I_2 = 0 \end{cases}$$

节点电压法:

$$\begin{cases} \frac{V_a - V_s}{R_1} + \frac{V_a}{R_2} + \frac{V_a - V_b}{R_3} = 0 \\ \frac{V_b - V_a}{R_3} + \frac{V_b}{R_4} + I_s = 0 \end{cases}$$

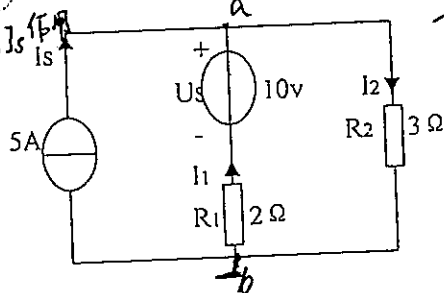
$$\begin{cases} \frac{V_a - 6}{2} + \frac{V_a}{2} + \frac{V_a - V_b}{1} = 0 \\ V_b - V_a + V_b + 3 = 0 \end{cases}$$

$$\begin{cases} 4V_a - 2V_b = 6 \\ -V_a + 2V_b + 3 = 0 \end{cases}$$

$$V_b = -1V, V_a = 3V$$

6. 用叠加原理和节点电压法分别求解下图中的电流 I_1 、 I_2 。

叠加: ① 单独 I_s 作用
 U_s 短路
② 单独 U_s 作用
 I_s 断路



叠加:

$$I_1 = -\frac{3}{2+3} \times 5 + \frac{10}{2+3} = -3 + 2 = -1A$$

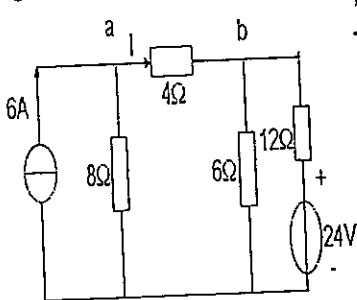
$$I_2 = \frac{2}{2+3} \times 5 + \frac{10}{2+3} = 2 + 2 = 4A$$

节点电压法:

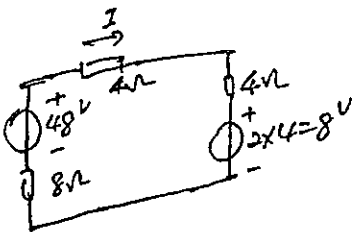
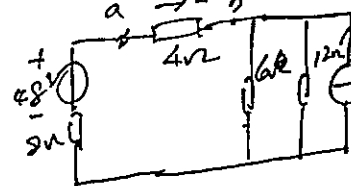
$$\frac{V_a - 10}{2} + \frac{V_a}{3} - 5 = 0 \Rightarrow V_a = \frac{60}{5} = 12V$$

$$\therefore I_2 = \frac{V_a}{3} = 4A, I_1 = -\frac{12 - 10}{2} = -1A$$

7. 分别用电源等效变换法、叠加原理、戴维南定理求电流 I 。



等效变换:



$$\therefore I = \frac{48 - 8}{8 + 4 + 4} = \frac{40}{16} = 2.5A$$

叠加:

$$I = \frac{8}{8+8} \times 6 - \frac{20}{12+6+12} \times \frac{6}{6+2+4} = 3 - \frac{20}{16} \times \frac{6}{18} = 3 - 0.5 = 2.5A$$

戴维南:

$$V_{ab0} = 6 \times 8 - \frac{20}{6+12} \times 6 = 48 - 8 = 40V$$

$$R_0 = 8 + 6 \parallel 12 = 8 + 4 = 12\Omega$$

$$\therefore I = \frac{V_{ab0}}{R_0 + 4} = \frac{40}{12 + 4} = \frac{40}{16} = 2.5A$$

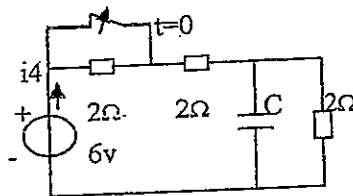
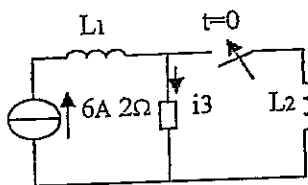
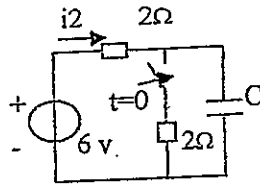
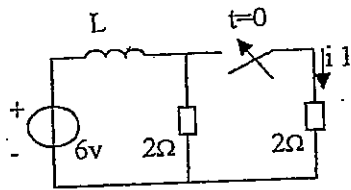
电路的过渡过程 选择题

李长泰

班级

姓名

图示各电路在换路前都已处于稳态，求换路后其中的电流 i 的初始值 $i(0+)$ 和稳态值 $i(\infty)$ 。



$$(1) \hat{u}_{L(0-)} = 6V = 3A = \hat{i}_{L(0+)}$$

$$\therefore \hat{i}_{1(0+)} = \frac{1}{2} \times 3 = 1.5A$$

$$\hat{u}_{L(\infty)} = 6V = 6A$$

$$\hat{i}_{1(\infty)} = \frac{6}{2} = 3A$$

$$(2) \hat{u}_{C(0-)} = 6V = \hat{u}_{C(0+)}$$

$$\hat{i}_{2(0+)} = \frac{6-6}{2} = 0A$$

$$\hat{i}_{2(\infty)} = \frac{6}{4} = 1.5A$$

$$\hat{u}_{C(\infty)} = \frac{1}{2} \times 6 = 3V$$

$$(3) \hat{i}_{L1(0+)} = \hat{i}_{L1(0-)} = 6A$$

$$\hat{i}_{L2(0-)} = 0 = \hat{i}_{L2(0+)}$$

$$\therefore \hat{i}_{3(0+)} = 6 - \hat{i}_{L2(0+)} = 6A$$

$$\hat{i}_{3(\infty)} = 0$$

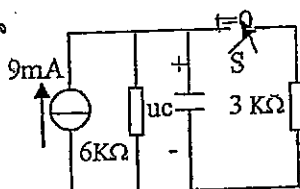
$$(4) \hat{u}_{C(0-)} = \frac{6}{2+2} \times 2 = 3V = \hat{u}_{C(0+)}$$

$$\hat{i}_{4(0+)} = \frac{6 - \hat{u}_{C(0+)}}{2+2} = \frac{6-3}{2+2} = \frac{3}{4}A$$

$$\hat{i}_{4(\infty)} = \frac{6}{2+2+2} = 1A$$

二、图示电路在换路前已处于稳态，求开关闭合后的电压 u_C ($C=2\mu f$)。

① 怎么求 R_0 。



$$u_{C(0+)} = u_{C(0-)} = 9 \times 6 = 54V$$

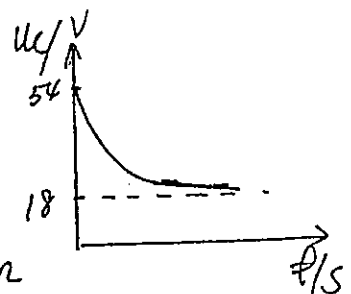
$$u_{C(\infty)} = 9 \times (3 \parallel 6) = 18V$$

$$\tau = R_0 C \quad R_0 = 3 \parallel 6 = 2k\Omega$$

$$\therefore \tau = 2 \times 10^3 \times 2 \times 10^{-6} = 4 \times 10^{-3} s$$

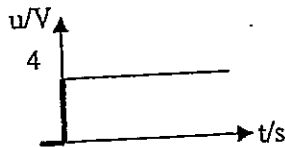
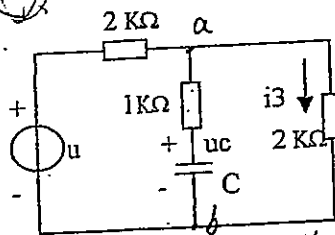
$$\therefore u_C(t) = u_{C(\infty)} + [u_{C(0+)} - u_{C(\infty)}] e^{-\frac{t}{\tau}} = 18 + [54 - 18] e^{-\frac{t}{4 \times 10^{-3}}}$$

$$= 18 + 36e^{-250t} V.$$



三、图示电路， u 为一阶跃电压如图所示，求 i_3 和 u_C 。设 $u_C(0^-) = 1V$, $C = 1\mu F$ 。

① 求 $U_{ab} =$
② 求 $R_0 =$



$$u_C(t) = u_C(0^-) = 1V$$

$$U_{ab}(t) = \frac{\frac{4}{2} + \frac{1}{1}}{\frac{1}{2} + 1 + \frac{1}{2}} = 1.5V$$

$$i_3(t) = \frac{U_{ab}(t)}{2} = 0.75mA$$

$$u_C(\infty) = \frac{4}{2} = 2V$$

$$i_3(\infty) = \frac{4}{2+2} = 1mA$$

$$R_0 = 1 + 2 \parallel 2 = 2k\Omega$$

$$\tau = R_0 C = 2 \times 10^3 \times 1 \times 10^{-6} = 2 \times 10^{-3}s$$

$$u_C(t) = 2 + [1 - 2]e^{-\frac{t}{2 \times 10^{-3}}}$$

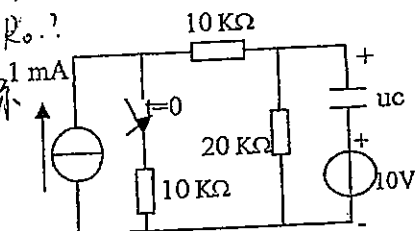
$$= 2 - e^{-500t} V$$

$$i_3(t) = 1 + (0.75 - 1)e^{-500t}$$

$$= 1 - 0.25e^{-500t} mA$$

四、图示电路在换路前已处于稳态，求开关闭合后的电压 u_C ($C = 10\mu F$)。变化曲线 $\Rightarrow$ NB: 绝对没单位。

怎么求 R_0 ?
把 u_C 看成电压源
往外看电路。



$$u_C(t) = 20 \times 1 - 10 = 10V$$

$$u_C(\infty) = \frac{10}{10+30} \times 20 - 10 = -5V$$

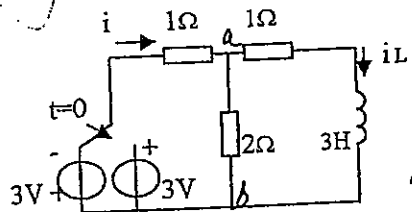
$$R_0 = 20 \parallel (10+10) = 10k\Omega$$

$$\tau = R_0 C = 10 \times 10^3 \times 10 \times 10^{-6} = 10^{-1}$$

$$u_C(t) = -5 + [10 + 5]e^{-10t}$$

$$= -5 + 15e^{-10t} V$$

五、图示电路在换路前已处于稳态，将开关从 1 合到 2 后求 i 和 i_L ，并画出变化曲线。



$$i_L(0^-) = i(0^-) \cdot \frac{2}{3} \quad i(0^-) = -\frac{3}{1+\frac{2}{3}} = -1.8A$$

$$i_L(0^+) = -1.8 \times \frac{2}{3} = -1.2A = i_L(t)$$

$$U_{ab}(t) = \frac{\frac{3}{1+\frac{1}{2}} - i_L(t)}{\frac{1}{1+\frac{1}{2}}} = \frac{3+1.2}{\frac{3}{2}} = 2.8V$$

$$i(t) = \frac{3 - U_{ab}(t)}{1} = 0.2A$$

$$i_L(\infty) = 1.2A$$

$$i(\infty) = \frac{3}{1+\frac{2}{3}} = 1.8A$$

$$R_0 = 1 + 1 \parallel 2 = \frac{5}{3}\Omega$$

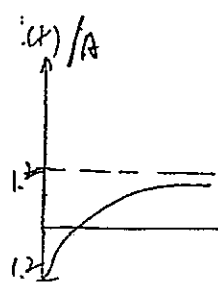
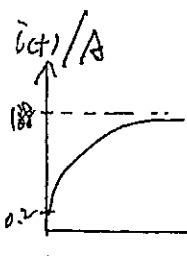
$$\tau = \frac{L}{R_0} = \frac{3}{\frac{5}{3}} = \frac{9}{5}s$$

$$i(t) = 1.8 + [0.2 - 1.8]e^{-\frac{5}{9}t}$$

$$= 1.8 - 1.6e^{-\frac{5}{9}t} A$$

$$i_L(t) = 1.2 + [-1.2 - 1.2]e^{-\frac{5}{9}t}$$

$$= 1.2 - 2.4e^{-\frac{5}{9}t} A$$



正弦交流电路 考大题(1道)

姓名

学号

一、日光灯管与整流器串联接到交流电压上,看作 R、L 串联。已知灯管电阻 $R_1=280\Omega$, 整流器电阻、电感分别为: $R_2=20\Omega$, $L=1.65H$ 。电源电压 $U=220V$, 求: 电路中的电流、灯管两端、整流器两端电压。电源频率 $50Hz$ 。

$$X_L = 2\pi fL = 314 \times 1.65 = 518 \Omega \quad |Z| = \sqrt{(280+20)^2 + 518^2} = 598.6 \Omega$$

$$\therefore I = \frac{U}{|Z|} = \frac{220}{598.6} = 0.368 A$$

$$\therefore U_R = IR = 0.368 \times 280 = 103 V$$

$$|Z_1| = \sqrt{20^2 + 518^2} = 518 \Omega$$

$$\therefore U_L = I|Z_1| = 0.368 \times 518 = 190.8 V$$

二、无源二端网络输入电压和电流为:

$$u = 220\sqrt{2}\sin(314t + 20^\circ) V$$

$$i = 4.4\sqrt{2}\sin(314t - 33^\circ) A \quad f = 50 Hz$$

求: 此二端网络由两个元件串联的等效电路和两个元件的参数值 并求网络的功率因数

1、输入有功 P、无功功率 Q

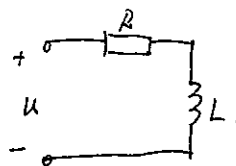
$$\text{求 } \varphi \Rightarrow Z \Rightarrow Z = \frac{U}{I} = \frac{220 \angle 20^\circ}{4.4 \angle -33^\circ} = 50 \angle 53^\circ = 30 + j40 = R + jX_L$$

$$\therefore R = 30 \Omega, X_L = 40 \Omega, \therefore L = \frac{X_L}{2\pi f} = \frac{40}{314} = 127 mH$$

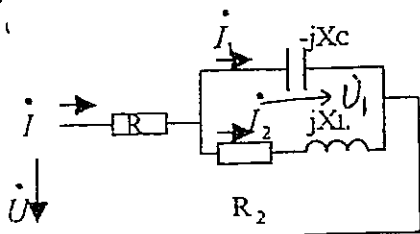
$$\varphi = 53^\circ \therefore \cos \varphi = \cos 53^\circ = 0.6$$

$$P = UI \cos \varphi = 220 \times 4.4 \times 0.6 = 580.8 W$$

$$Q = UI \sin \varphi = 220 \times 4.4 \times 0.8 = 774.4 VAR$$



三、图中, $I_1=10A$, $I_2=10\sqrt{2}A$, $U=220V$, $R=5\Omega$, $R_2=X_L$, 求: I , R_2 , X_C , X_L 。



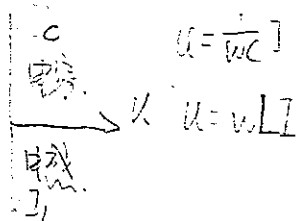
$$\vec{U}_1 = U \angle 0^\circ \quad \vec{I}_1 = 10 \angle -45^\circ \quad \vec{I}_2 = 10\sqrt{2} \angle 45^\circ$$

$$\therefore \vec{I} = \vec{I}_1 + \vec{I}_2 = 10 \angle -45^\circ + 10\sqrt{2} \angle 45^\circ = 10 \angle -45^\circ + 10 \angle 45^\circ = 20 \angle 0^\circ A$$

$$\vec{U}_R = 220 - \vec{I}R = 220 - 10 \times 5 = 170 V$$

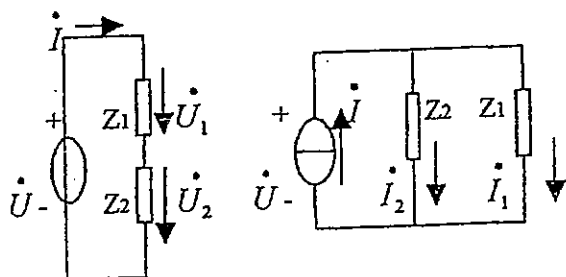
$$\therefore X_C = \frac{U_R}{I_1} = \frac{170}{10} = 17 \Omega$$

$$X_L = R = \frac{170}{10\sqrt{2}} \cos 45^\circ = \frac{17}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2} = 8.5 \Omega$$

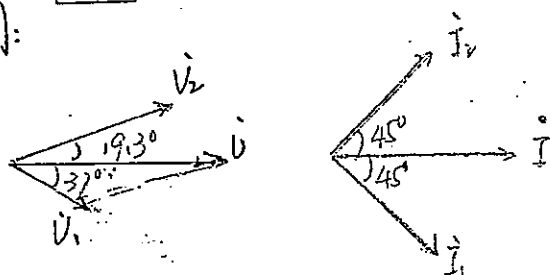


四、(a) 求电流 $\dot{I}$ 和各阻抗上的电压 $\dot{U}_1$ 与 $\dot{U}_2$ ，并作相量图。 $\dot{U}=10\angle 0^\circ \text{ V}$ 、 $Z_1=2\Omega$ 、 $Z_2=2+j3\Omega$ 。

(b) 求图中各支路电流 $\dot{I}_1$ 与 $\dot{I}_2$ 和电压 $\dot{U}$ ，并作相量图。 $\dot{I}=2\angle 0^\circ \text{ A}$ 、 $Z_1=1+j1\Omega$ 、 $Z_2=1-j1\Omega$



相量图:



$$(a) \quad Z_2=2+j3=3.6\angle 56.3^\circ \quad \dot{I}=\frac{\dot{U}}{Z_1+Z_2}=\frac{10}{5\angle 37^\circ}=2\angle -37^\circ$$

$$\dot{U}_1=\frac{Z_1}{Z_1+Z_2}\cdot\dot{U}=\frac{2}{4+j3}\cdot 10\angle 0^\circ=\frac{2}{5\angle 37^\circ}\cdot 10\angle 0^\circ$$

$$=4\angle -37^\circ$$

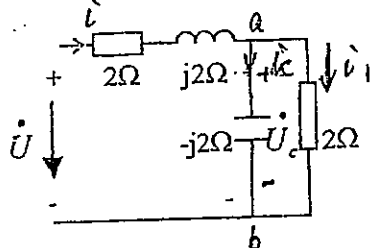
$$\dot{U}_2=\frac{Z_2}{Z_1+Z_2}\cdot\dot{U}=\frac{3.6\angle 56.3^\circ}{5\angle 37^\circ}\cdot 10\angle 0^\circ=7.2\angle 19.3^\circ \text{ V}$$

$$(b) \quad \dot{I}_1=\frac{Z_2}{Z_1+Z_2}\cdot\dot{I}=\frac{1-j1}{2}\times 2\angle 0^\circ=\sqrt{2}\angle -45^\circ \text{ (A)}$$

$$\dot{I}_2=\frac{Z_1}{Z_1+Z_2}\cdot\dot{I}=\frac{1+j1}{2}\times 2\angle 0^\circ=\sqrt{2}\angle 45^\circ \text{ (A)}$$

$$\dot{U}=\dot{I}Z_1=\dot{I}_2Z_2=\sqrt{2}\angle 45^\circ\cdot\sqrt{2}\angle 45^\circ=2\angle 90^\circ \text{ (V)}$$

五、图示电路， $\dot{U}_c=1\angle 0^\circ \text{ V}$ ，求： $\dot{U}$



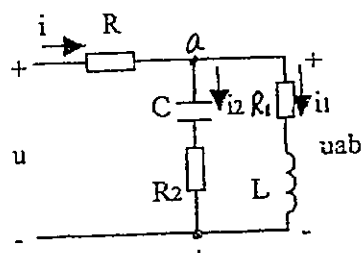
假设为正向如图， $Z_{ab}=\frac{-j2\times 2}{-j2+2}=\frac{-j4}{2-j2}=\sqrt{2}\angle -45^\circ=1-j$

$$\therefore \dot{U}=\frac{\dot{U}_c}{Z_{ab}}\cdot 2$$

$$=\frac{1\angle 0^\circ}{\sqrt{2}\angle -45^\circ}\cdot\sqrt{10}\angle 18.4^\circ$$

$$=\sqrt{5}\angle 63.4^\circ \text{ (V)}$$

六、已知： $R=R_1=R_2=10\Omega$ 、 $L=31.8\text{mH}$ 、 $C=318\mu\text{F}$ 、 $f=50\text{Hz}$ 、 $U=10\text{V}$ 。求并联支路电压 U_{ab} 及电路的 P 、 Q 、 S 及 $\cos\varphi$ 。



$$X_L=2\pi fL=314\times 31.8\times 10^{-3}=10\Omega$$

$$X_C=\frac{1}{2\pi fC}=\frac{1}{314\times 318\times 10^{-6}}=10\Omega$$

$$Z_{ab}=(10+j10)\parallel(10-j10)=\frac{10\sqrt{2}\angle 45^\circ\cdot 10\sqrt{2}\angle -45^\circ}{20}$$

$$=10\Omega$$

$$\therefore U_{ab}=\frac{1}{2}U=5\text{ V}$$

$$|Z_{ab}|=\sqrt{R^2+X^2}=10\sqrt{2}=14.14\Omega$$

$$X_C=\frac{1}{2\pi fC}=\frac{1}{2\pi\times 50\times 318\times 10^{-6}}=10\Omega$$

$$X_L=2\pi fL=2\pi\times 50\times 31.8\times 10^{-3}=10\Omega$$

$$|Z_{ab}|=\frac{1}{\frac{1}{10}+\frac{1}{10}}=5\Omega$$

$$\cos\varphi=1$$

$$\varphi=0^\circ$$

$$P=UI=\frac{10}{20}\times 10=5\text{ W}=5\text{ (VA)}$$

$$Q=0$$

$$U_{ab}=\frac{U}{R+|Z_1||Z_2|}\times |Z_1||Z_2|$$

七、今有 40W 日光灯一个，使用时灯管（纯电阻）与整流器（纯电感）串联接在电压为 220V、频率 50 Hz 的电源上，灯管两端电压 110V，求整流器感抗与电感、此时的功率因数。若将功率因数提高到 0.8，应并多大电容。

$$\frac{1}{R_{XJ}} = \frac{P_{XJ}}{U_{XJ}^2} = \frac{40}{110^2} = 0.0033 \quad R_{XJ} = 302.5 \Omega \quad I = \frac{U_{XJ}}{R_{XJ}} = \frac{110}{302.5} = 0.364 \text{ A}$$

$$|Z| = \frac{U}{I} = \frac{220}{0.364} = 605 \quad \therefore X_L = \sqrt{|Z|^2 - R_{XJ}^2} = 524 \Omega \quad L = \frac{X_L}{\omega} = \frac{524}{314} = 1.67 \text{ H}$$

$$\cos \varphi_1 = \frac{R_{XJ}}{\sqrt{R_{XJ}^2 + X_L^2}} = \frac{302.5}{605} = 0.5 \quad \varphi_1 = 60^\circ$$

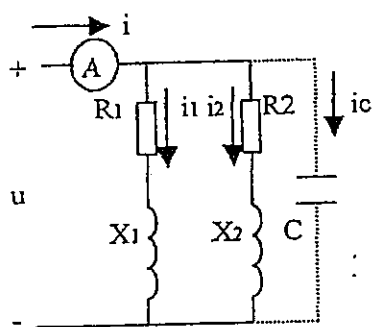
$$\cos \varphi = 0.8 \quad \varphi = 36.9^\circ$$

$$C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi) = \frac{40}{314 \times 220^2} [\tan 60^\circ - \tan 36.9^\circ]$$

$$= 2.63 \times 10^{-6} [1.732 - 0.75] = 2.58 \times 10^{-6} \text{ F}$$

$$= 2.58 \mu\text{F}$$

八、图中， $U=220\text{V}$ 、 $f=50\text{Hz}$ 、 $R_1=10\Omega$ 、 $X_1=10\sqrt{3}\Omega$ 、 $R_2=5\Omega$ 、 $X_2=5\sqrt{3}\Omega$ 。(1) 求电流表读数 I 及 $\cos \varphi$ (2) 欲使电路功率因数提高到 0.866 需并多大的电容？(3) 并联电容后的电流表的读数为多少。



$$Z_1 = R_1 + jX_1 = 10 + j10\sqrt{3} = 20 \angle 60^\circ$$

$$Z_2 = R_2 + jX_2 = 5 + j5\sqrt{3} = 10 \angle 60^\circ$$

$$Z_1 || Z_2 = \frac{20}{3} \angle 60^\circ$$

$$(1) \therefore I = \frac{U}{|Z|} = \frac{3}{20} \times 220 = 33 \text{ A}$$

$$\cos 60^\circ = \frac{1}{2} \quad P = UI \cos \varphi = 220 \times 33 \times \frac{1}{2} = 3630 \text{ W}$$

$$\cos \varphi = 0.866 \quad \varphi = \arccos 0.866 = 30^\circ$$

$$(2) \therefore C = \frac{P}{\omega U^2} [\tan \varphi_1 - \tan \varphi]$$

$$= \frac{3630}{314 \times 220^2} [\tan 60^\circ - \tan 30^\circ]$$

$$= 238.8 \times 10^{-6} [1.732 - 0.577]$$

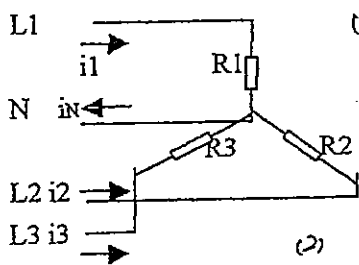
$$= 276 \times 10^{-6} \text{ F}$$

$$(3) I' = \frac{P}{U \cos \varphi} = \frac{3630}{220 \times 0.866} = 19.1 \text{ A}$$

三相 选择填空

姓名 学号 考卷号

一、图示三相四线制电路，电源线电压 $U_L=380V$ ， $R_1=11\Omega$ 、 $R_2=R_3=22\Omega$ 。求：(1) 负载相电压、相电流及中线电流，并作它们的相量图；(2) 如无中线，求负载相电压及中性点电压；(3) 如无中线，当 L_1 相短路时各相电压和电流；(4) 如无中线， L_3 相断路时求另外两相的电压和电流；(5) 如有中线，则 (3)、(4) 情况如何？



(1) 设 $\dot{U}_A = 220\angle 0^\circ$, 则 $\dot{U}_B = 220\angle -120^\circ$, $\dot{U}_C = 220\angle 120^\circ$

$$\dot{I}_A = \frac{220}{11} = 20A \quad \dot{I}_B = \frac{220\angle -120^\circ}{22} = 10\angle -120^\circ \quad \dot{I}_C = \frac{220\angle 120^\circ}{22} = 10\angle 120^\circ$$

$$\dot{I}_N = \dot{I}_A + \dot{I}_B + \dot{I}_C = 20 + 10\angle -120^\circ + 10\angle 120^\circ = 20 + 2 \times 10 \cos 60^\circ = 40A$$

(2) $\dot{U}_{NN'} = \frac{\frac{\dot{U}_A}{R_1} + \frac{\dot{U}_B}{R_2} + \frac{\dot{U}_C}{R_3}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} = \frac{\frac{220}{11} + \frac{220\angle -120^\circ}{22} + \frac{220\angle 120^\circ}{22}}{\frac{1}{11} + \frac{1}{22} + \frac{1}{22}} = 55V$

$\therefore U_A' = 220 - 55 = 165V$ $\dot{U}_B = 220\angle -120^\circ - 55 = 251\angle 131^\circ \therefore U_B = 251 = U_C$

(3) 无中线， L_1 相短路， R_2 和 R_3 接在火线电压上。相电压 = 线电压 = $\frac{380}{\sqrt{3}} = 219V$

A 相电流 (线) $\dot{I}_A = \frac{380\angle 0^\circ}{22} + \frac{380\angle 180^\circ}{22} = \frac{380}{22} [1 + \cos 60^\circ + j \sin 60^\circ] = \frac{190}{11} (3 + j)$

$$= 30\angle 30^\circ \therefore I_A = 30A$$

(4) 无中线 A 相断路，则另外两相串联接在 BC 之间，A 相电压 = $\frac{1}{3}$ 线电压 = $127V$

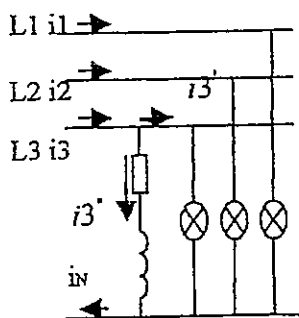
$$I = \frac{380}{33} = 11.5A \quad U_B' = \frac{2}{3} U_L = 253V$$

(5) 如果有中线，(3) L_1 相短路断开，其余两相不受影响。(4) L_3 线电流为 0 其余两相不变

二、图示电路，三相四线制电源电压 $380/220V$ ，接有对称星接白炽灯负载，总功率 $180W$ 。

此外，在 L_3 相上接一额定电压为 $220V$ 、 $40W$ 、 $\cos\varphi=0.5$ 的日光灯一只，求： $\dot{I}_1, \dot{I}_2, \dot{I}_3, \dot{I}_N$

设 $\dot{U}_1 = 220\angle 0^\circ V$ 。



对称负载 $P = 5 U_L I_L \therefore I_L = \frac{P}{\sqrt{3} U_L} = \frac{180}{\sqrt{3} \times 380} = 0.273A$

$$\therefore \dot{I}_A = 0.273\angle 0^\circ \quad \dot{I}_B = 0.273\angle -120^\circ \quad \text{日光灯: } I = \frac{P}{U \cos\varphi} = \frac{40}{220 \times 0.5} = 0.364A$$

$\dot{I}_C = 0.273\angle 120^\circ + 0.364\angle 120^\circ - 60^\circ$

$$= -0.1365 + j0.2364 + j0.182 + j0.3152$$

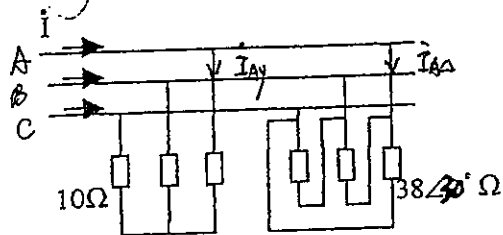
$$= 0.0455 + j0.5516 = 0.553\angle 85.3^\circ A$$

$\dot{I}_N = \dot{I}_A + \dot{I}_B + \dot{I}_C = 0.273\angle 0^\circ + 0.273\angle -120^\circ + 0.553\angle 85.3^\circ = 0.364\angle 60^\circ A$

$= \dot{I}_3'$

$\therefore \dot{I}_3' + \dot{I}_1 + \dot{I}_2 = 0$ (j 线电流, 对称负载)

三、线电压 380V，接有两组对称负载，求电流 I 及总的 P、Q。



以 $\dot{I}_A$ 为参考

Y接: $\dot{V}_A = 220\angle 0^\circ$ $\dot{V}_{AB} = 380\angle 30^\circ$

$$\dot{I}_{Ay} = \frac{220\angle 0^\circ}{10} = 22 \text{ A}$$

$$\dot{I}_{A\Delta(\Delta)} = \frac{380\angle 30^\circ}{38\angle 30^\circ} = 10 \text{ A}$$

$$\dot{I}_{A\Delta} = 10\sqrt{3}\angle -30^\circ \quad \therefore \dot{I}_A = \dot{I}_{Ay} + \dot{I}_{A\Delta}$$

$$\therefore \dot{I}_A = 22 + 10\sqrt{3}\angle -30^\circ = 22 + 10\sqrt{3}\cos 30^\circ - j10\sqrt{3}\sin 30^\circ = 22 + 15 - j8.66 = 37 - j8.66$$

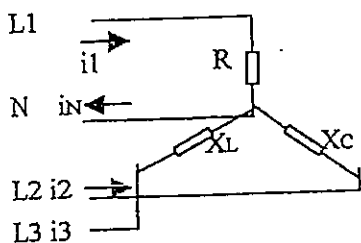
$$= 38\angle -13.2^\circ \text{ A}$$

总有功: $P = P_2 + P_y = \sqrt{3} \times 380 \times 10\sqrt{3} \cdot \cos 30^\circ + \sqrt{3} \times 380 \times 22 = 8872 + 14480 = 24352 \text{ W}$

$$Q = \sqrt{3} \times 380 \times 10\sqrt{3} \sin 30^\circ = 5700 \text{ Var}$$

$$Q_y + Q_\Delta = Q_\Delta = 0$$

四、图示三相四线制电路，电源线电压 $U_L = 380\text{V}$ ， $R = X_L = X_C = 10\Omega$ 。问 (1) 负载是否对称？
(2) 求各相电流及中线电流 (3) 求三相平均功率 P。



(1) 不对称. 相位差 120°

$$(2) \dot{V}_A = 220\angle 0^\circ \quad \dot{V}_B = 220\angle -120^\circ \quad \dot{V}_C = 220\angle 120^\circ$$

$$\dot{I}_1 = \frac{220}{R} = 22 \text{ A} \quad \dot{I}_2 = \frac{220\angle -120^\circ}{-jX_C} = 22\angle 30^\circ$$

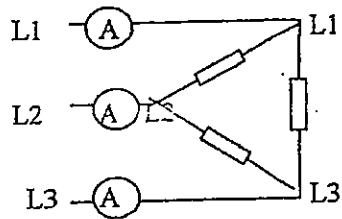
$$\dot{I}_3 = 22\angle 30^\circ$$

$$\therefore \dot{I}_N = 22 + 22\angle -30^\circ + 22\angle 30^\circ = 22 + 2 \times 22 \cos 30^\circ = 22 + 22\sqrt{3} = 60.1 \text{ A}$$

(3) $*P = \sqrt{3} U_L I_L \cos \phi$ 不能用. 因为负载不对称

$$P = I_1^2 R = 22^2 \times 10 = 4840 \text{ W}$$

五、对称负载角接，，电源电压 220V，电流表读数 17.3A，三相功率 $P=4.5\text{KW}$ ，求：(1) 每相负载的电阻和感抗，(2) 当 L1、L2 间的相断开时，图中各电流表读数和总功率 P ，(3) 当 L1 线断开时图中各电流表读数和总功率 P 。



$$(1) P = \sqrt{3} U_e I_e \cos \varphi \quad \cos \varphi = \frac{P}{\sqrt{3} U_e I_e} = \frac{4.5 \times 10^3}{\sqrt{3} \times 220 \times 17.3} = \frac{15}{22}$$

$$|Z| = \frac{U_p}{I_p} = \frac{220}{10} = 22$$

$$R = |Z| \cos \varphi = 15 \Omega \quad X_L = |Z| \sin \varphi = \sqrt{22^2 - 15^2} = 16 \Omega$$

$$(2) L_1 \text{ 和 } L_2 \text{ 间断开, } I_c \text{ 不变, } I_A = I_B = I_p = \frac{17.3}{\sqrt{3}} = 10 \text{ A}$$

$$P' = \frac{2}{3} P = \frac{2}{3} \times 4.5 = 3 \text{ kW}$$

$$(3) L_1 \text{ 线断开, } I_A = 0$$

$$I_B = I_C = \frac{U_{BC}}{|Z|} + \frac{U_{BC}}{2|Z|} = \frac{U_{BC}}{|Z|} \cdot \frac{3}{2} = \frac{220}{22} \times \frac{3}{2} = 15 \text{ A}$$

$$P'' = \frac{1}{3} P + \frac{1}{2} \times \frac{1}{3} P = 1.5 + 0.75 = 2.25 \text{ kW}$$

↓ 电阻减小一半，电功率增大一倍

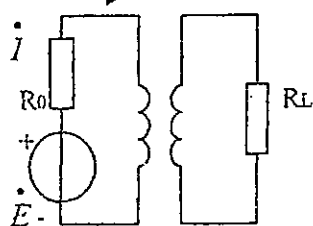
一、单相变压器，容量 10KVA、电压 3300/220V。今欲在二次绕组上接 60W220V 的白炽灯，要求变压器在额定状态下运行，问可以接多少个灯？一、二次绕组的额定电流分别为多少？

$$n = \frac{10 \times 10^3}{60} = 166.7$$

$$I_{N1} = \frac{S_N}{U_{N1}} = \frac{10000}{3300} = 3.03 \text{ A}$$

$$I_{N2} = \frac{S_N}{U_{N2}} = \frac{10000}{220} = 45.5 \text{ A}$$

二、中， $R_L = 8\Omega$ ，已知 $N_1 = 300$ 、 $N_2 = 100$ 信号源电动势 $E = 6\text{V}$ ，内阻 $R_0 = 100\Omega$ 。求信号源输出的功率。



$$k = \frac{300}{100} = 3, R_L' = k^2 \cdot R_L = 9 \times 8 = 72 \Omega$$

$$P = \left(\frac{6}{100 + 72} \right)^2 \times 72 = 87 \text{ mW}$$

三、已知 Y100L1-4 型异步电动机额定数据如下：2.2KW 380V Y 接，1420r/min $\cos\phi = 0.82$ $\eta = 80\%$ 。求：(1) 额定相电流、线电流、负载转矩 (2) 额定转差率及额定负载时转子电流频率。电源频率 50HZ。

$$(1) P_1 = \sqrt{3} U_L I_L \cos\phi$$

$$P_2 = P_1 \cdot \eta$$

$$\therefore I_L = \frac{P_2}{\eta \cdot \sqrt{3} U_L \cos\phi} = \frac{2.2 \times 10^3}{0.8 \times \sqrt{3} \times 380 \times 0.82} = 5.09 \text{ A}$$

$$P = \sqrt{3} U_L I_L \cos\phi$$

$$P = 3 U_p I_p \cos\phi$$

$$T_N = 9550 \cdot \frac{P_N}{n_N} = 9550 \cdot \frac{2.2}{1420} = 14.8 \text{ N}\cdot\text{m}$$

$$(2) s_N = \frac{1500 - 1420}{1500} = 5.3\%$$

$$f_2 = s_N \cdot f_1 = 2.67 \text{ Hz}$$

四、某四极三相异步电动机额定功率为 30KW，额定电压为 380V，三角形连接，频率 50HZ。在额定负载下运行是其转差率 0.02，效率 90%，线电流 57.5A，求：(1) 转子旋转磁场对转子的转速；(2) 额定转矩；(3) 电动机功率因数；(4) 如果 $T_{st}/T_N=1.2$ ， $I_{st}/I_N=7$ 用 Y-Δ 换接启动时的启动电流和启动转矩；(5) 当负载转矩为额定转矩的 60% 和 25% 时能否启动？

$$(1) \Delta n = 0.02 \times 1500 = 30 \text{ r/min}$$

$$n = 1500 - 1500 \times 0.02 = 1470 \text{ r/min}$$

$$(2) T_N = 9550 \cdot \frac{30}{1470} = 194.9 \text{ N}\cdot\text{m}$$

$$(3) \cos \varphi = \frac{P_2}{\sqrt{3} U_L I_L \eta} = \frac{30 \times 10^3}{\sqrt{3} \times 380 \times 57.5 \times 0.9} = 0.88$$

$$(4) T_{st} = 1.2 T_N = 233.88 \text{ N}\cdot\text{m}$$

$$T_{stY-\Delta} = \frac{1}{3} T_{st} = 77.96 \text{ N}\cdot\text{m}$$

$$I_{st} = 7 I_N = 7 \times 57.5 = 402.5 \text{ A}$$

$$I_{stY-\Delta} = \frac{1}{3} \times 402.5 = 134.2 \text{ A}$$

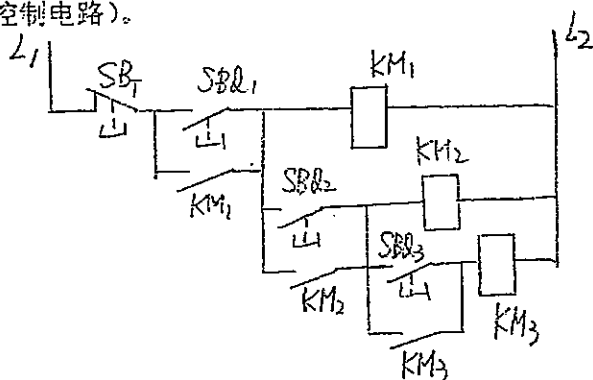
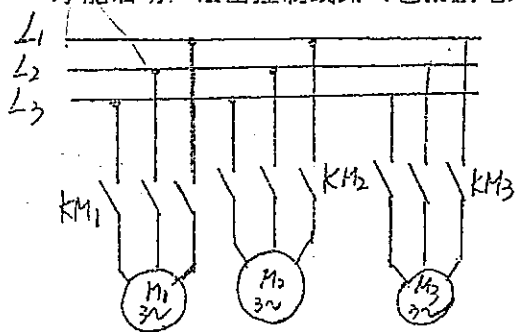
(5) 当 $T_2 = 60\% T_N$ 时，

$$T_{stY-\Delta} = \frac{1}{3} \times 1.2 T_N = 40\% T_N$$

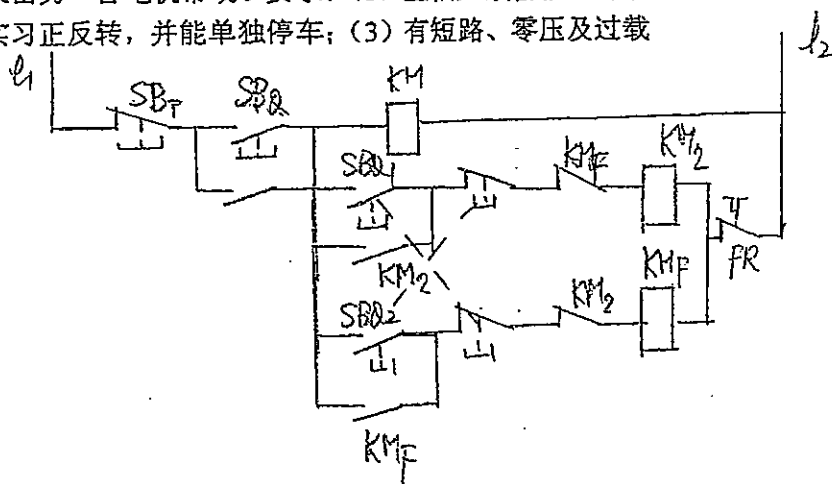
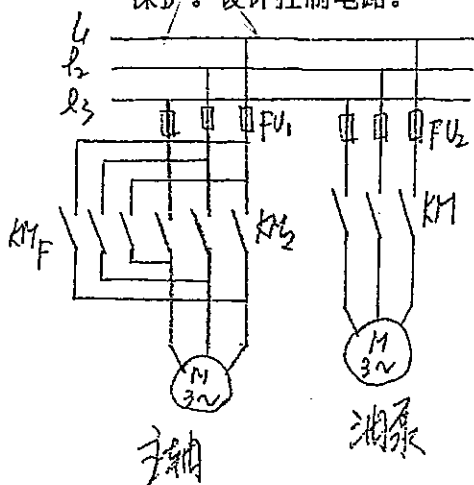
∴ 不能启动

$T_2 = 25\%$ 时可以启动

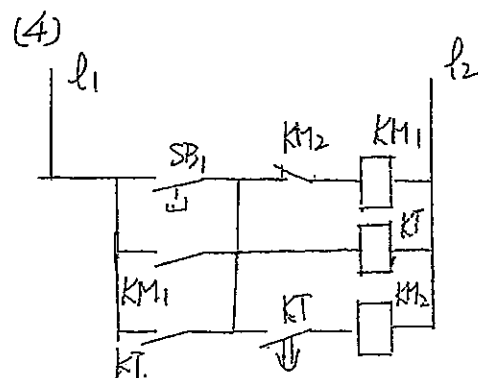
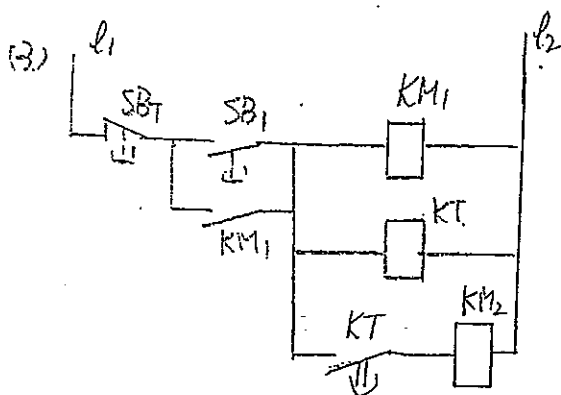
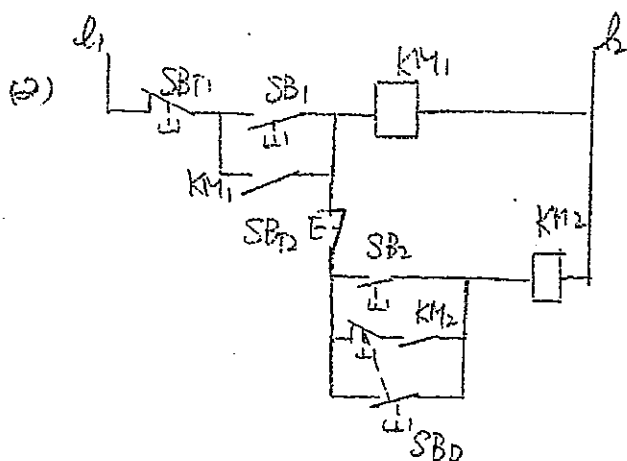
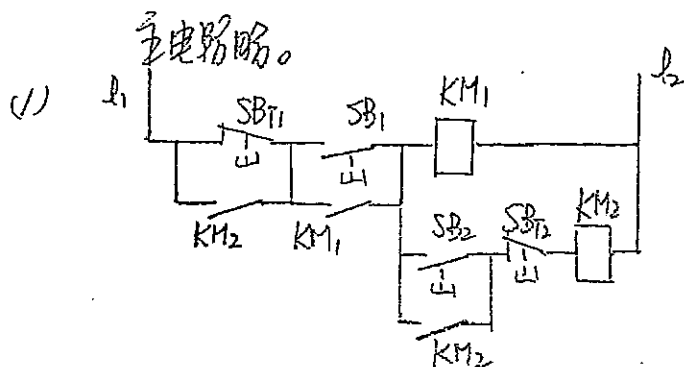
五、今要求三台电动机 M1、M2、M3 顺序启动，M1 启动后 M2 才能启动、M2 启动后 M3 才能启动，画出控制线路（包括主电路和控制电路）。



六、某机床主轴由一台电机带动，油泵由另一台电机带动。要求：(1) 主轴必须在油泵开动后才能开动；(2) 主轴要求能用电器实现正反转，并能单独停车；(3) 有短路、零压及过载保护。设计控制电路。



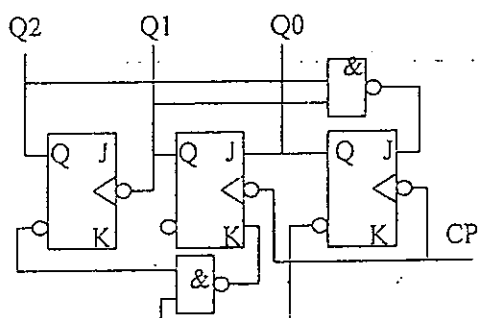
七、根据以下五个条件分别绘制控制电路 (M1、M2 都是笼型电机): (1) M1 起动后 M2 才能起动, M2 能单独停车; (2) M1 起动后 M2 才能起动, M2 能点动; (3) M1 先起动, 经一段时间后 M2 自行起动; (4) M1 先起动, 经一段时间后 M2 自行起动, M2 启动后 M1 立即停车; (5) 起动时, M1 起动后 M2 才能起动; 停止时 M2 停止后 M1 才能停止。



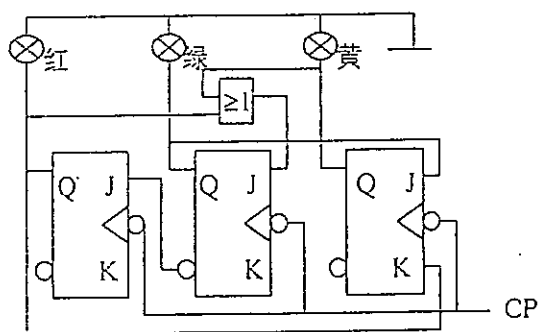
(5) 在图(1)中加上联锁按钮

~~十二、~~ 用 74LS161 同步计数器接成十二进制计数器：(1) 用清零法；(2) 用置数法。

十三、 列出下图计数状态表，说明它是几进制计数器。初态为 000。



十四、 下图，输出 1 时灯可以被点亮，分析彩灯点亮的顺序，初态为 000。



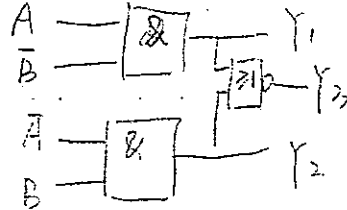
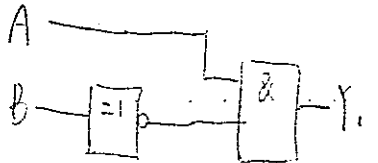
八、有两个数字的比较器，逻辑关系如下表，写出输出表达式并画出逻辑电路。

输入		输出		
A	B	Y1(A>B)	Y2(A<B)	Y3(A=B)
0	0	0	0	0
0	1	0	1	0
1	0	1	0	0
1	1	0	0	1

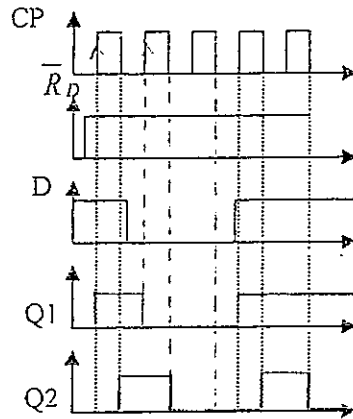
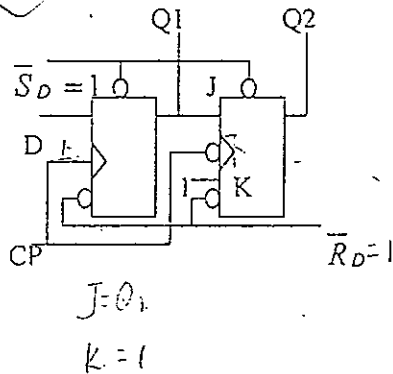
$$Y_1 = A\bar{B}$$

$$Y_2 = \bar{A}B$$

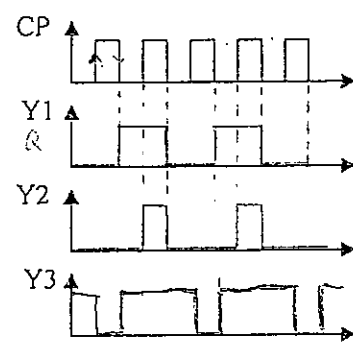
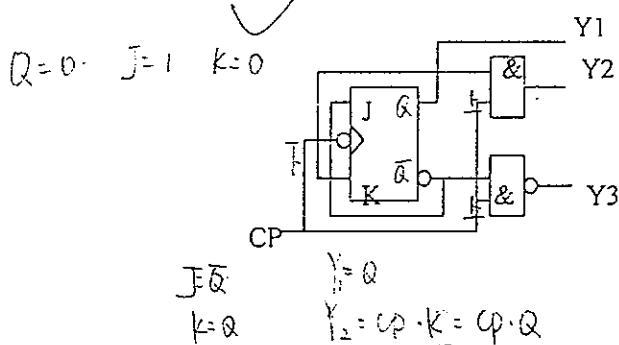
$$Y_3 = AB + \bar{A}\bar{B} = \overline{Y_1 + Y_2}$$



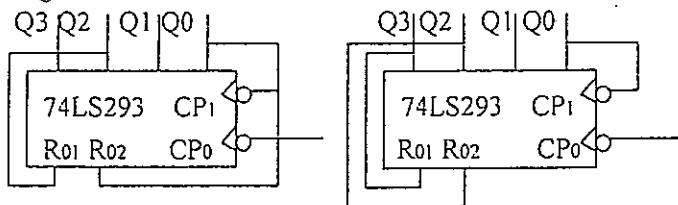
九、根据逻辑电路及相应的输入波形画出 Q1 和 Q2 端的输出波形。初态 Q1=Q2=0



十、画出 Y1、Y2、Y3 的波形，触发器初态为 0。



图示电路各为几进制？如何用它接成七进制计数器。



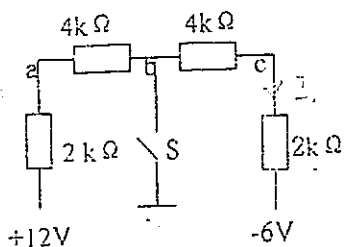
直流电路

班级

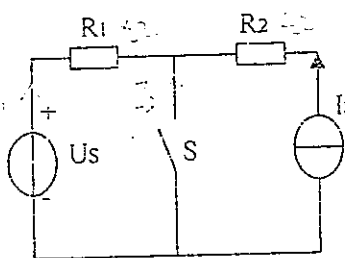
姓名

学号

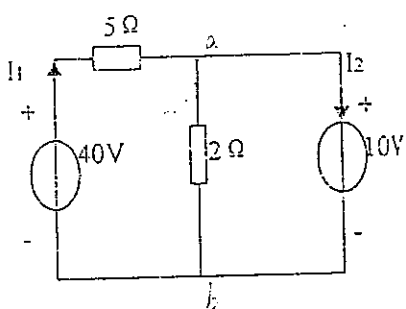
- 1、求图示电路中开关 S 闭合和断开两种情况下 a、b、c 三点的电位。



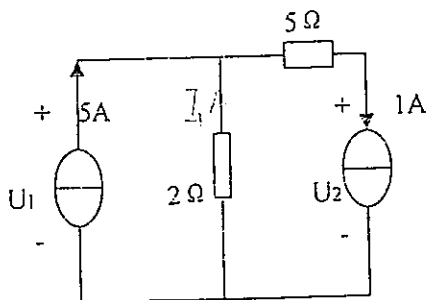
- 2、求图示电路中，已知 $U_s=6V$ ， $I_s=2A$ ， $R_1=R_2=4\Omega$ 。求：开关 S 闭合时通过开关的电流和开关 S 断开时两端的电压（在图中注明所选择的参考方向）。



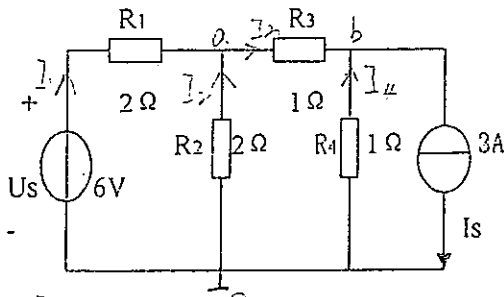
- 3、求图中通过恒压源的电流 I_1 、 I_2 及其功率，并说明是起电源作用还是负载作用。



- 4、求图中恒流源两端的电压 U_1 、 U_2 及其功率，并说明是起电源作用还是负载作用。



5、用支路电流法和节点电压法分别求解下图中各支路电流（在图中注明所选择的参考方向）



(1) 节点电压法

由 KCL 定律可知 $I_1 + I_2 = I_3$

$$I_1 + I_2 = I_3 = 3A \quad \text{①}$$

$$U_{ac} = (I_2 - I_3)R_1 = 0 \quad \text{②}$$

$$U_{bc} = -I_4 R_4 = 0 \quad \text{③}$$

$$U_{bc} = I_3 R_3 - I_4 R_4 = I_3 R_3 - 0 = I_3 R_3 = 0 \quad \text{④}$$

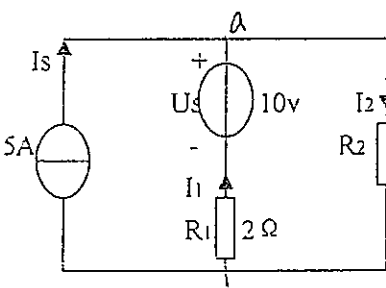
由 ①②③④ 得 $I_1 = 3.5A, I_2 = 1.5A$

$$I_3 = 3A, I_4 = 1A$$

$$a. \begin{cases} I_1 + I_2 = I_3 \\ I_3 + I_4 = I_s \end{cases}$$

$$\begin{cases} \frac{V_s - V_a}{R_1} + \frac{0 - V_a}{R_2} = \frac{V_a - V_b}{R_3} \\ \frac{V_a - V_b}{R_3} + \frac{0 - V_b}{R_4} = I_s \end{cases}$$

6、用叠加原理和节点电压法分别求解下图中的电流 I_1 、 I_2 。



$$I_s + I_1 = I_2$$

$$I_s + \frac{V_a - V_s}{R_1} = \frac{V_a}{R_2}$$

$$I_1 = 1A$$

$$I_2 = 4A$$

$$U_1 = \frac{R_1}{R_1 + R_2} U_s = 4V$$

$$U_2 = \frac{R_2}{R_1 + R_2} U_s = 6V$$

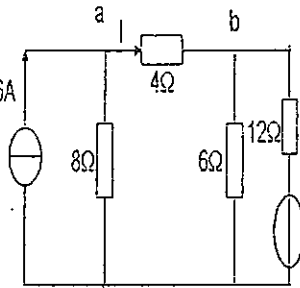
$$I_1 = \frac{U_1}{R_1} = 1A$$

$$I_2 = \frac{U_2}{R_2} = 2A$$

$$I_2 = I_1 + I_s = 4A$$

$$I_1 = I_2 - I_s = 1A$$

7、分别用电源等效变换法、叠加原理、戴维南定理求电流 I 。

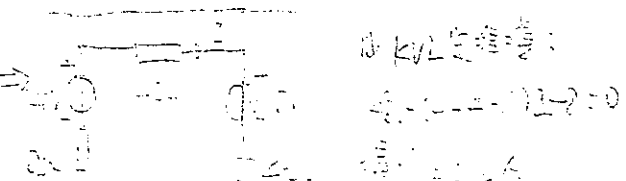
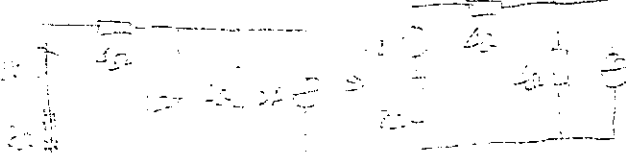


$$I = \frac{V_a - V_b}{4}$$

$$\frac{V_a}{4} + \frac{V_a - V_b}{4} - 6 = 0$$

$$\frac{V_a - V_b}{4} - \frac{V_b}{6} = \frac{V_b - 24}{12}$$

(1) 电源等效变换法



叠加原理：①分解的电路画出来
②再求解。

② 叠加原理

活电路求 I

$$I = \frac{V_a - V_b}{4} = 2A$$

$$I = \frac{V_a - V_b}{4} = 2A$$

$$I = \frac{V_a - V_b}{4} = 2A$$

$$I = \frac{V_a - V_b}{4} = 2A$$

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$$I = \frac{V_a - V_b}{4} = 2A$$

③ 戴维南定理

求开路电压

$$U_{oc} = 6 \times 8 = 48V$$

$$V_b = \frac{1}{12} \times 24 = 2V$$

$$I = \frac{U_{oc} - V_b}{4 + 8} = 2A$$

$$I = \frac{U_{oc} - V_b}{4 + 8} = 2A$$

$$I = \frac{U_{oc} - V_b}{4 + 8} = 2A$$

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$$I = \frac{U_{oc} - V_b}{4 + 8} = 2A$$

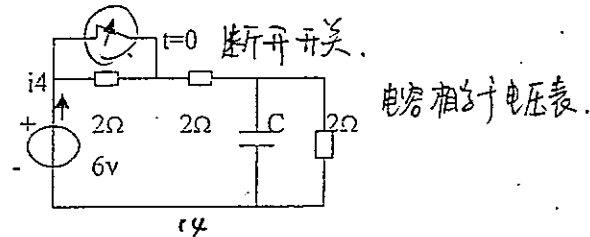
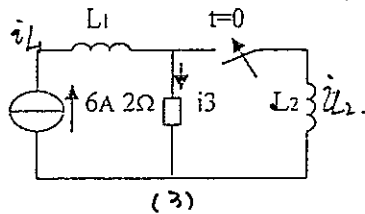
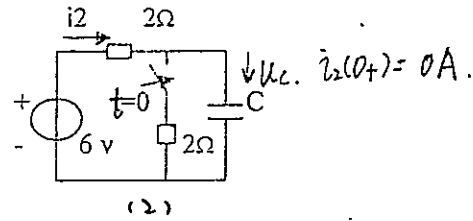
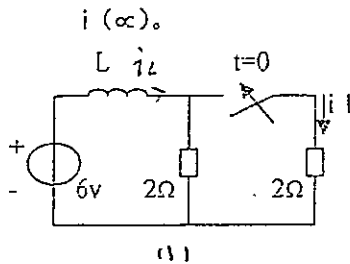
电路暂态分析

班级 1304

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一、图示各电路在换路前都已处于稳态，求换路后其中的电流 i 的初始值 $i(0+)$ 和稳态值 $i(\infty)$ 。



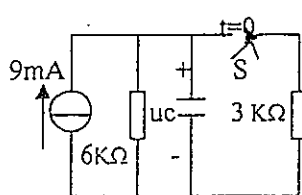
$$\begin{aligned} (1) \quad i_L(0-) &= \frac{6}{2} = 3A \\ i_L(0+) &= i_L(0-) = 3A \\ i(0+) &= \frac{2}{2+2} \cdot 3 = 1.5A \\ i(\infty) &= \frac{6}{2} = 3A \end{aligned}$$

$$\begin{aligned} (2) \quad U_C(0-) &= 6V \\ U_C(0+) &= U_C(0-) = 6V \\ i_2(0+) &= \frac{6-6}{2} = 0A \\ i_2(\infty) &= \frac{6}{2+2} = 1.5A \end{aligned}$$

$$\begin{aligned} (3) \quad i_{L1}(0+) &= i_{L1}(0-) = 6A \\ i_{L2}(0+) &= i_{L2}(0-) = 0A \\ i_3(0+) &= i_{L1}(0+) - i_{L2}(0+) = 6A \\ i_3(\infty) &= 0A \end{aligned}$$

$$\begin{aligned} (4) \quad U_C(0+) &= U_C(0-) = \frac{2}{2+2} \times 6 = 3V \\ i_4(0+) &= \frac{U_C(0+)}{2\Omega} = 1.5A \quad i_4(0+) = \frac{6-3}{2+2} = 0.75A \\ i_4(\infty) &= \frac{6}{2+2+2} = 1A \end{aligned}$$

二、图示电路在换路前已处于稳态，求开关闭合后的电压 u_C ($C=2\mu f$)。



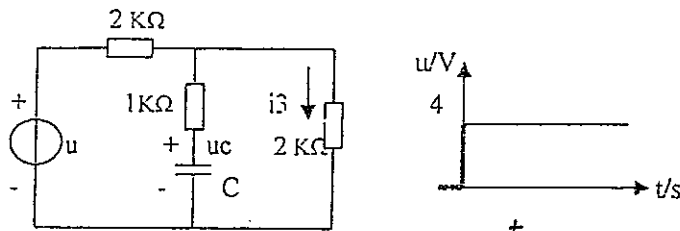
$$\begin{aligned} U_C(0-) &= 9 \times 10^{-3} \times 6 \times 10^3 = 54V \\ U_C(0+) &= U_C(0-) = 54V \\ U_C(\infty) &= \frac{3 \times 10^3}{3 \times 10^3 + 6 \times 10^3} \times 9 \times 10^{-3} \times 6 \times 10^3 = 18V \end{aligned}$$

$$R = 6 \parallel 3 = 2k\Omega$$

$$\tau = RC = 2\mu f \times 2k\Omega = 4 \times 10^{-3} s$$

$$\begin{aligned} U_C &= U_C(\infty) + (U_C(0+) - U_C(\infty)) e^{-\frac{t}{\tau}} \\ &= 18 + (54 - 18) e^{-\frac{t}{4 \times 10^{-3}}} \\ &= 18 + 36 e^{-250t} V \end{aligned}$$

三、图示电路， u 为一阶跃电压如图所示，求 i_3 和 u_c 。设 $u_c(0^-) = 1V$ ， $C = 1\mu f$ 。



$$\text{则有 } u_c = u_c(\infty) + (u_c(0^+) - u_c(\infty))e^{-\frac{t}{\tau}}$$

$$= 2 + (1 - 2)e^{-\frac{t}{2 \times 10^{-3}}}$$

$$= 2 - e^{-500t} \text{ V}$$

$$i_c = C \frac{du_c}{dt} = 1 \times 10^{-6} \times (500e^{-500t}) = 5 \times 10^{-4} e^{-500t} \text{ A}$$

$$i_3 = \frac{u_c + 1 \times 10^3 i_c}{2 \times 10^3} = (1 - 0.25e^{-500t}) \times 10^{-3} \text{ A}$$

$$u_c(0^-) = u_c(0^+) = 1V$$

$$u_c(\infty) = \frac{2}{2+2} \times 4 = 2V$$

$$R = 1 + \frac{2 \times 2}{2+2} = 2k\Omega$$

$$\tau = RC = 2 \times 1\mu f = 2 \times 10^{-3} s$$

$$\text{写公式 } f(t) = f(\infty) + [f(0^+) - f(\infty)]e^{-\frac{t}{\tau}}$$

$$u_c(0^+) = u_c(0^-) = 1V$$

$$u_c(\infty) = \frac{1}{2}u = 2V$$

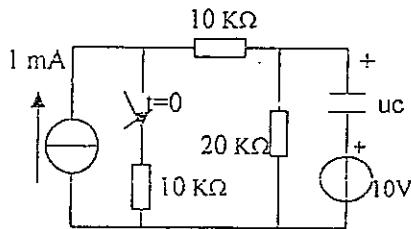
$$\frac{V_a(0^+) - u}{2} + \frac{V_a - u_c(0^+)}{1} + \frac{V_a(0^+)}{2} = 0$$

$$\therefore V_a(0^+) = 1.5V$$

$$\therefore i_3(0^+) = \frac{V_a(0^+)}{2} = 0.75 \text{ mA}$$

$$i_3(\infty) = \frac{u}{2} = 1 \text{ mA}$$

四、图示电路在换路前已处于稳态，求开关闭合后的电压 u_c 并画出曲线 ($C = 10\mu f$)。



$$u_c(0^+) = u_c(0^-) = 1 \text{ mA} \times 20k\Omega = 20V$$

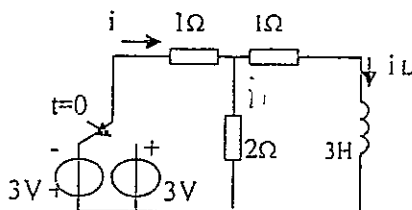
$$u_c(\infty) = \frac{10k\Omega}{(10+20+10)k\Omega} \times 1 \text{ mA} \times 20k\Omega = 5V$$

$$\tau = R_o C = [10 \parallel 10 \parallel 20] \times 10\mu f = 0.1s$$

$$u_c = u_c(\infty) + (u_c(0^+) - u_c(\infty))e^{-\frac{t}{\tau}}$$

$$= 5 + 15e^{-10t} \text{ V}$$

五、图示电路在换路前已处于稳态，将开关从 1 合到 2 后求 i 和 i_L ，并画出变化曲线。



$$i_L(0^+) = i_L(0^-) = \frac{1 \parallel 2}{1 + (1 \parallel 2)} \times 3 = 1.2 \text{ A}$$

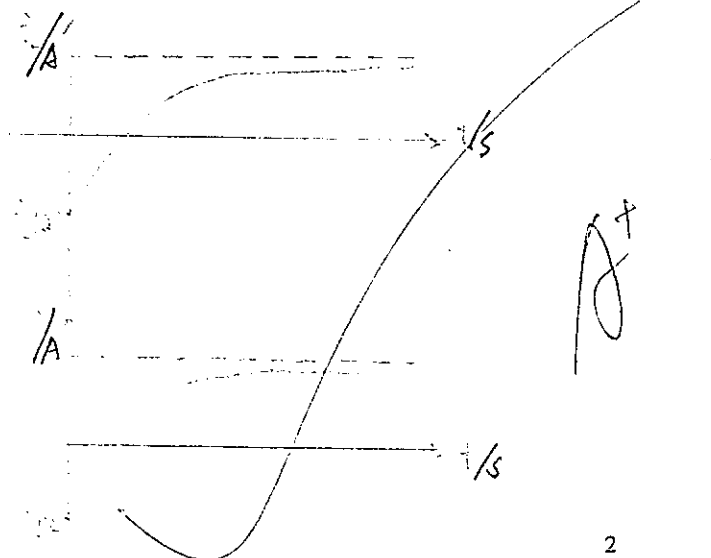
$$i_L(\infty) = 1.2 \text{ A}$$

$$\tau = \frac{L}{R_o} = \frac{3}{1 + (1 \parallel 2)} = 1.8s$$

$$i_L = i_L(\infty) + (i_L(0^+) - i_L(\infty))e^{-\frac{t}{\tau}}$$

$$= 1.2 + 2.4e^{-\frac{5}{9}t} \text{ A} \quad i_L = \frac{2}{2+1}i$$

$$i = \frac{i_L \times 1}{2} = 0.6 - 1.2e^{-\frac{5}{9}t} \text{ A} \Rightarrow i = \frac{3}{2}i_L$$



正弦交流电路和三相电路

姓名

班号学号

- 一、日光灯管与整流器串联接到交流电压上，看作 R、L 串联。已知灯管电阻 $R_1=280\Omega$ ，整流器电阻、电感分别为： $R_2=20\Omega$ ， $L=1.65H$ 。电源电压 $U=220V$ ，求：电路中的电流、灯管两端、整流器两端电压。电源频率 50Hz。

$$\omega = 2\pi f = 100\pi = 314 \text{ rad/s}$$

$$\text{设电源电压 } u = 220\sqrt{2}\sin(314t + 0^\circ) \quad \dot{U} = 220\angle 0^\circ \text{ V}$$

$$X_L = \omega L = 314 \times 1.65 = 518.15 \Omega$$

$$Z = R_2 + jX_L + R_1 = 300 + 518.1j = 598.7\angle 60^\circ \Omega$$

$$\dot{I} = \frac{\dot{U}}{|Z|} = \frac{220\angle 0^\circ}{598.7\angle 60^\circ} = 0.368\angle -60^\circ \text{ A}$$

$$\text{则 } i = 0.368\sqrt{2}\sin(314t - 60^\circ)$$

$$U_1 = R_1 \dot{I} = 0.368\angle -60^\circ \times 280 = 103 \text{ V}$$

$$U_2 = R_2 \dot{I} = 518 \times 0.368 = 190.8 \text{ V}$$

- 二、无源二端网络输入电压和电流为：
 $u = 220\sqrt{2}\sin(314t + 20^\circ) \text{ V}$
 $i = 4.4\sqrt{2}\sin(314t - 33^\circ) \text{ A}$

求：此二端网络由两个元件串联的等效电路和两个元件的参数值，并求网络的功率因数 λ 、输入有功功率 P、无功功率 Q。

$$\dot{U} = 220\angle 20^\circ \text{ V}, \quad \dot{I} = 4.4\angle -33^\circ \text{ A}$$

$$Z = \frac{\dot{U}}{\dot{I}} = 50\angle 53^\circ = 30 + j40 \Omega$$

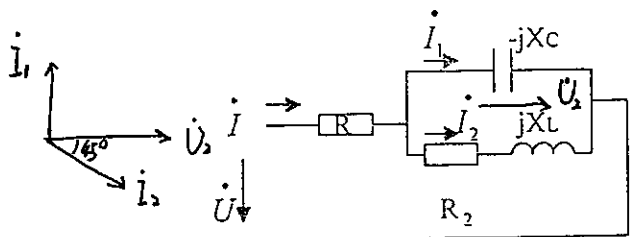
$$X_L = \omega L = 40 \Omega \quad \therefore L = 127 \text{ mH}$$

$$P = UI \cos \varphi = 220 \times 4.4 \cos 53^\circ = 580.8 \text{ W}$$

$$Q = UI \sin \varphi = 220 \times 4.4 \sin 53^\circ = 774.4 \text{ var}$$

$$\lambda = \cos \varphi = \frac{3}{5}$$

- 三、图中， $I_1=10\text{A}$ ， $I_2=10\sqrt{2}\text{A}$ ， $U=220\text{V}$ ， $R=5\Omega$ ， $R_2=X_L$ ，求： I 、 R_2 、 X_C 、 X_L 。



$$\dot{U}_2 = U_2\angle 0^\circ \text{ V}, \quad \dot{I}_1 = 10\angle 90^\circ = j10 \text{ A}$$

$$R_2 = X_L, \quad \tan \frac{\varphi_2}{2} = 1$$

$$\dot{I}_2 = 10\sqrt{2}\angle -45^\circ = 10 - j10 \text{ A}$$

$$\dot{I} = \dot{I}_1 + \dot{I}_2 = j10 + 10 - j10 = 10 \text{ A}$$

$$I = 10 \text{ A}$$

$$\dot{U}_3 = \dot{U} - \dot{I}R \quad \dot{U}_R = \dot{I}R = 10 \times 5 = 50 \text{ V}$$

$$\therefore \dot{U}_R \dot{U}_2 \dot{U} \Rightarrow \text{同相}$$

$$U_2 = 220 - 50 \text{ V} = 170 \text{ V}$$

$$X_C = \frac{U_2}{I_1} = 17 \Omega \quad |Z| = \frac{U_2}{I_2} = 17 \Omega$$

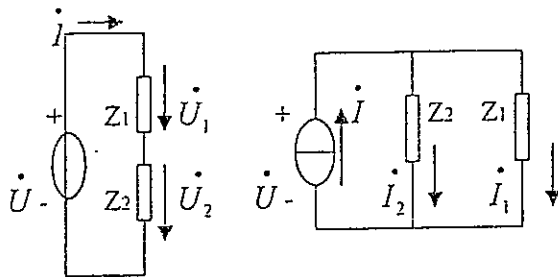
$$\lambda = \cos \varphi = \frac{3}{5}$$

$$Z_2 = \sqrt{2}R_2\angle 45^\circ$$

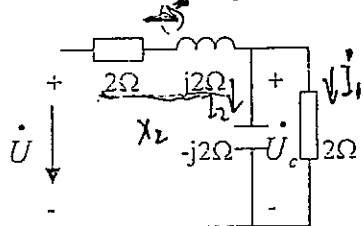
$$|Z_2| = \sqrt{2}R$$

四、(a) 求电流 $\dot{I}$ 和各阻抗上的电压 $\dot{U}_1$ 与 $\dot{U}_2$ ，并作相量图。 $\dot{U} = 10\angle 0^\circ \text{V}$ 、 $Z_1 = 2\Omega$ 、 $Z_2 = 2 + j3\Omega$ 。

(b) 求图中各支路电流 $\dot{I}_1$ 与 $\dot{I}_2$ 和电压 $\dot{U}$ ，并作相量图。 $\dot{I} = 2\angle 0^\circ \text{A}$ 、 $Z_1 = 1 + j1\Omega$ 、 $Z_2 = 1 - j1\Omega$



五、图示电路 $\dot{U}_C = 1\angle 0^\circ \text{V}$ ，求： $\dot{U}$



$$\dot{I}_1 = \frac{\dot{U}_C}{R} = 0.5\angle 0^\circ \text{A}$$

$$\dot{I}_2 = \frac{\dot{U}_C}{X_C} = \frac{1\angle 0^\circ}{2\angle 90^\circ} = 0.5\angle 90^\circ$$

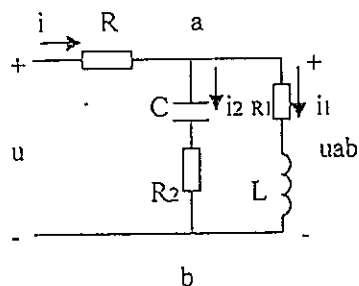
$$\dot{I} = \dot{I}_1 + \dot{I}_2 = 0.5\angle 45^\circ$$

$$X_2 = 2 + j2 = 2\sqrt{2}\angle 45^\circ$$

$$\dot{U}_2 = \dot{I} X_2 = \sqrt{2}\angle 90^\circ = j\sqrt{2}$$

$$\therefore \dot{U} = \dot{U}_2 + \dot{U}_C = 1 + j\sqrt{2} = \sqrt{5}\angle 49^\circ \text{V} \quad \tan \varphi = 2$$

六、已知： $R = R_1 = R_2 = 10\Omega$ ， $L = 31.8\text{mH}$ ， $C = 318\mu\text{F}$ ， $f = 50\text{Hz}$ ， $U = 10\text{V}$ 。求并联支路电压 U_{ab} 及电路的 P 、 Q 、 S 及 $\cos\varphi$ 。



$$X_L = \omega L = 2\pi f L = 10\Omega$$

$$X_C = \frac{1}{\omega C} = 10\Omega$$

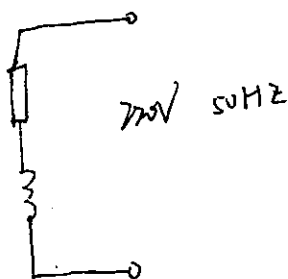
$$Z_{ab} = (R_2 - jX_C) \parallel (R_1 + jX_L)$$

$$= \frac{10\angle -45^\circ \times 10\angle 45^\circ}{10 - j10 + 10 + j10}$$

$$= 10\Omega$$

$$U_{ab} = \frac{10}{10 + j10} U = 5\text{V}$$

七、今有 40W 日光灯一个，使用时灯管（纯电阻）与整流器（纯电感）串联接在电压为 220V、频率 50 Hz 的电源上，灯管两端电压 110V，求整流器感抗与电感、此时的功率因数。若将功率因数提高到 0.8，应并多大电容。



$$R = \frac{U^2}{P} = \frac{110^2}{40} = 302.5 \Omega$$

$$I = \frac{P}{U} = \frac{40}{220} = 0.1818 \text{ A}$$

$$X_L = \frac{U}{I} = \frac{110}{0.1818} \Omega = 605 \Omega$$

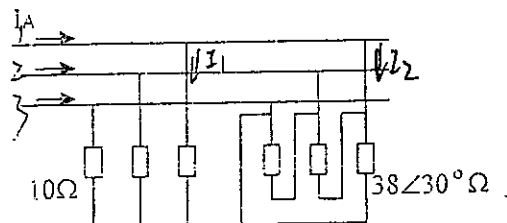
$$X_L = 2\pi fL$$

$$L = \frac{X_L}{2\pi f} = \frac{605}{2\pi \times 50} = 9.7 \text{ mH}$$

$$\cos\phi = \frac{R}{Z}$$

$$C =$$

八、线电压 380V，接有两组对称负载，求电流 I_A 及总的 P、Q。



$$I_1 = \frac{U_{12}}{R} = \frac{380}{10} = 38 \text{ A}$$

$$I_2 = 0$$

$$\dot{U}_{12} = 380 \angle 30^\circ \text{ V}$$

$$\dot{U}_1 = 220 \angle 0^\circ \text{ V}$$

$$I_1 = \frac{\dot{U}_1}{R} = 22 \angle 0^\circ \text{ A}$$

$$I_2 = \frac{\dot{U}_{12}}{R} = 38 \angle -30^\circ \text{ A}$$